
Course Name: Machine Learning for Earth System Sciences

Week 05 - Assignment 05 - (Jul-2026)

TYPE OF QUESTIONS: MCQ/MSQ

Number of questions: 10

Total marks: 10x1= 10

QUESTION 1

Which of the following statements is/are True about the study - Shaby BA, Reich BJ, Cooley D, Kaufman CG. *A Markov-switching model for heat waves. The Annals of Applied Statistics*. 2016 Mar;10(1):74-93.

- I. The study employs a 2 - state Bayesian latent state model for modelling heat waves with interpretability as one of the objectives.
- II. In the case studies of Moscow and Paris, only maximum daily temperatures have been considered, the inclusion of daily minimum temperature might had been of more interests for climate analysts.
- III. One of the drawbacks of the proposed model is that the model also classifies short-lived extreme high temperature periods as it doesn't contain duration in its classification criteria.

- (A) I and II
- (B) I and III
- (C) II and III
- (D) All

Answer: A

Detailed Solution:

Refer to the mentioned research paper

QUESTION 2

Which of the following is the most appropriate one-line summary of the work - Katzfuss M, Hammerling D, Smith RL. *A Bayesian hierarchical model for climate change detection and attribution. Geophysical Research Letters*. 2017 Jun 16;44(11):5720-8.

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- (A) The paper presents a Bayesian hierarchical approach to the study of climate change problem which has been conventionally attempted by Regression based methods in the past.
- (B) The paper presents a Bayesian hierarchical model that addresses the open questions in regression-based climate change detection and attribution, by taking into account uncertainties related due to observations, external forcings and climate variability.
- (C) The paper presents a Regression based model for detecting climate change and attribution incorporating various uncertainties in observations, missing data and climate variability in the absence of any forcing
- (D) The paper presents a Bayesian hierarchical model for detecting climate change and attribution by formulating climate change as a multi variate spatial or spatio-temporal regression problem including the uncertainties in observations.

Answer: B

Detailed Solution:

Refer to the mentioned research paper

QUESTION 3

Which of the following features makes Capsule Networks (CapsNets) a better choice over Convolutional Networks (ConvNets) for a data driven framework for extreme-causing weather patterns.

- I. CapsNets use pooling layers thus runs faster than ConvNets.
- II. CapsNet seek invariance rather than equivariance.
- III. CapsNets are better than ConvNets for capturing the spatial relationships between key features.

(A) II

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- (B) I and II
- (C) II and III
- (D) III

Answer: D

Detailed Solution:

CapsNets can capture the spatial relationships between various features better than ConvNets and CapsNet uses capsule instead of pooling layers.

QUESTION 4

Refer to the study - *Sharma A, Mitra A, Vasan V, Govindarajan R. Spatio-temporal relationships between rainfall and convective clouds during Indian monsoon through a discrete lens. International Journal of Climatology. 2021 Feb;41(2):1351-68.*

In this work OLR has been used as one of the proxies to measure convective cloud cover and intuitively OLR should have a negative correlation with precipitation, but in some regions of the landmass of India, during monsoon there is no rainfall in spite of low OLR. Which regions show this type of variations in the correlation between OLR and precipitation?

- (A) North Eastern India
- (B) Eastern Peninsular India
- (C) Peninsular India
- (D) Peninsular India except the west coast

Answer: D

Detailed Solution:

Refer to the mentioned research paper

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QUESTION 5

In the study - Geng YA, Li Q, Lin T, Jiang L, Xu L, Zheng D, Yao W, Lyu W, Zhang Y. *Lightnet: A dual spatiotemporal encoder network model for lightning prediction. In Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining 2019 Jul 25 (pp. 2439-2447).*, the Lightnet model proposed for lightening prediction is based on

- (A) Simulation data from WRF
- (B) Weather Observational Data
- (C) Both simulation data and recent observational data
- (D) Reanalysis Data

Answer: C

Detailed Solution:

Refer to the mentioned research paper

QUESTION 6

In the study - Berrocal VJ, Guan Y, Muyskens A, Wang H, Reich BJ, Mulholland JA, Chang HH. *A comparison of statistical and machine learning methods for creating national daily maps of ambient PM_{2.5} concentration. Atmospheric Environment. 2020 Feb 1;222:117130.*, Statistical methods such as Universal Kriging have been reported to outperform machine learning methods for air pollution exposure assessment. Which of the following ML models could have been a better choice or more competitive against the statistical models for the task?

- (A) Logistic Regression
- (B) Convolution Neural Networks
- (C) LSTM
- (D) Decision Tree

Answer: B

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Detailed Solution:

Statistical methods have been reported to outperform Machine Learning methods because statistical methods account for spatial dependence. Thus, ML models that leverage spatial features can potentially be better competitors to statistical methods.

QUESTION 7

Match the following, in reference to the research papers reviewed in this course.

W: Analog Forecasting	P: Equivariance
X : OLR	Q: Look in the past
Y : CapsNet	R: Spatial Interpolation
Z: Kriging	S: Cloud Cover

(A) W-Q, X-S, Y-P, Z-R

(B) W-Q, X-S, Y-R, Z-P

(C) W-P, X-S, Y-R, Z-Q

(D) W-Q, X-R, Y-S, Z-P

Answer: A

Detailed Solution:

Analog forecasting - past data is referred for analogy/ matching pattern

OLR – used as a proxy for measuring convective cloud cover.

CapsNet – seek equivariance

Kriging - spatial interpolation technique used in environmental science

QUESTION 8

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In the model developed for the prediction of lightening described in the study - Geng YA, Li Q, Lin T, Jiang L, Xu L, Zheng D, Yao W, Lyu W, Zhang Y. Lightnet: A dual spatiotemporal encoder network model for lightning prediction. In Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining 2019 Jul 25 (pp. 2439-2447). The most appropriate statement describing the purpose of the fusion module is

- A. Combining WRF simulation data and recent observations to eliminate the deviations in time and space domain and the consequent biases for the prediction.
- B. Combining the WRF simulation data with a deep learning-based model output for better short-term predictions.
- C. Combining WRF features and recent observational features to improve short-term predictions
- D. Combining WRF features and recent observational features to improve short-term predictions and calibrate long-term predictions.

Answer: D

Detailed Solution:

The Fusion module in this work is used to fuse features from the WRF data and the recent observational data. This has been done because the errors in the NWP simulations demonstrate different patterns in different spatio-temporal locations, making it hard to calibrate the simulation.

QUESTION 9

Complete the following sentence in reference to the study - Mitra A, Apte A, Govindarajan R, Vasan V, Vadlamani S. A discrete view of the Indian monsoon to

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identify spatial patterns of rainfall. Dynamics and Statistics of the Climate System. 2018;3(1):dzy009.

The authors have employed ____ based approach to identify spatial patterns for Indian Summer Monsoon rainfall considering the daily rainfall data of total ____ days from the summer monsoon period months of 8 years over 357 locations and compared the results with K-means and ____ algorithms. The study has reported that only ____ spatial patterns can represent the spatial rainfall distribution for 95% days of each monsoon season.

- A. Graph, 122, Spectral clustering, 10
- B. Markov Random Field, 122, Spectral clustering, 8
- C. Markov Random Field, 976, Spectral clustering, 10
- D. Markov Random Field, 976, Gaussian Mixture Model, 8

Answer: C

Detailed Solution:

Markov Random Field approach has been used

Total days in data is 122 days (from 4 monsoon months) * 8 years (2000- 2007) = 976

The comparisons have been done with K-means and Spectral Clustering Algorithms

Total 10 spatial patterns have been identified in the study

QUESTION 10

In reference to the study - Kathuria D, Mohanty BP, Katzfuss M. A nonstationary geostatistical framework for soil moisture prediction in the presence of surface heterogeneity. Water Resources Research. 2019 Jan;55(1):729-53., which of the following statements is/are True?

- (I) In traditional geostatistical models, the spatial variance/correlation of soil moisture has been assumed to be constant and recent studies have found this assumption as true.

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(II) Surface characteristics like soil texture, topography and vegetation vary the spatial variance/correlation of soil moisture and the effect of vegetation is dynamic based on stage of crop, season and time of the year.

- (A) I only
- (B) II only
- (C) Both
- (D) None

Answer: B

Detailed Solution:

The assumption that variance/correlation of soil moisture is constant in traditional geostatistical studies has been found to be incorrect in recent studies, refer to the paper for references.